

Sungsoo Kim

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Research Interests

My current interests include **Scattering Amplitudes and Particle Physics**.

Publications

- [1] **The Diagrammar of Quantum Magnusian** May 2026
L. Guo^{*}, J. H. Kim^{*}, J. W. Kim^{*}, **S. S. Kim^{*}**, S. M. Lee^{*}, J. R. Li^{*}
arXiv:2605.25473 [hep-th]
- [2] **On the integrability of root-Kerr probe dynamics** Apr 2026
S. S. Kim^{*}, S. M. Lee^{*}
arXiv:2604.26544 [hep-th]
- [3] **Classical eikonal in relativistic scattering** Nov 2025
S. S. Kim^{*}, H. J. Lee^{*}, S. M. Lee^{*}
JHEP **11** (2025) 032
- [4] **Classical Eikonal from Magnus Expansion** Jan 2025
J. H. Kim^{*}, J. W. Kim^{*}, **S. S. Kim^{*}**, S. M. Lee^{*}
JHEP **01** (2025) 111
^{*}Equal contribution (co-first authors)

Experience

Completed Projects

- The Diagrammar of Quantum Magnusian** Ended May 2026
Collaborators: Li Guo Kim^{*}, Joon-Hwi Kim^{*}, Jung-Wook Kim^{*}, Sungsoo Kim^{*}, Sangmin Lee^{*}, Joon-Hwi Kim^{*}, Jian-Rong Li
- Developed a generalized diagrammatic framework for Quantum Magnusian structures and investigated systematic relations between graph topology and scattering observables.
 - Studied rooted and non-rooted graph structures and their associated combinatorial weights in Quantum Magnusian formulations.
 - Established computational rules for organizing generalized graph structures arising in Magnusian expansions.
 - Implemented the complete computational framework in *Mathematica* and developed the public package *Quantum-Magnusian* for automated graph generation and evaluation.
- On the integrability of root-Kerr probe dynamics** Ended Apr 2026
Collaborators: Sungsoo Kim^{*}, Sangmin Lee^{*}
- Investigated integrability properties of probe dynamics in root-Kerr systems, a non-gravitating limit of Kerr-Newman black holes.
 - Demonstrated that integrability persists to all spin orders at leading probe-charge order when interactions follow the Newman–Janis prescription.
 - Showed that at second order in probe charge the conservation can be extended to spin-squared order but breaks down at spin-cubic order.
 - Established through asymptotic conservation arguments that the spin-cubic obstruction cannot be restored by further deformations of the probe action.
- Classical Eikonal in Relativistic Scattering** Ended Aug 2025
Collaborators: Sungsoo Kim^{*}, Hojin Lee^{*}, Sangmin Lee^{*}
- Established an explicit diagrammatic correspondence between the Magnus expansion and WQFT, clarifying how non-rooted trees and causal weights arise in relativistic kinematics.
 - Analyzing rigorously the standard WQFT setup with two straight-line worldlines interacting by electromagnetic or gravitational fields.
 - Demonstrated that the eikonal framework uniformly treats particles and fields, enabling radiation observables—such as Compton and 1.5PM emission—to be derived from the same generating functional.
- Classical Eikonal from Magnus Expansion** Ended Nov 2024
Collaborators: Joon-Hwi Kim^{*}, Jung-Wook Kim^{*}, Sungsoo Kim^{*}, Sangmin Lee^{*}
- Reformulated the classical eikonal phase as a Magnus expansion of the time-evolution operator connecting in- and

out-states.

- Proved this expansion holds for all tree-level diagrams and identified the coefficient structure associated with directional propagators.
- Contributed to extending the Murua formula to general tree-level diagrams and clarifying its role in computing diagrammatic weights.
- Implemented the complete expansion algorithm in Mathematica and, **at the Wolfram team's suggestion, uploaded the example file to the Wolfram Community as an exemplar of research-grade Mathematica usage.**

*Equal contribution (co-first authors)

CERN Summer Student Programme 2023

Jun 2023 – Aug 2023

Summer Student, CERN

Project: *Experimental Analysis of Crucial Decays with Potential to Violate the Standard Model*

Supervisors: Chen Chen, Jinlin Fu

- **Project Paper:** Feasibility Investigation into Detecting $D_s^+ \rightarrow \tau^+ + \nu_\tau$ in $p\text{Ne}$ Collisions at LHCb [Link to Paper].
- **Presentation:** Presented findings at the LHCb Summer Student Presentation on August 29, 2023.
- Selected as **one of two representatives** from Korea for the fully funded CERN Summer Student Programme.
- Gained hands-on experience in experimental research within the LHCb experiment, including understanding current experimental processes and challenges.

Scholarships & Awards

Presidential Science Scholarship

Mar 2021 – Feb 2026

- Provided full tuition and additional benefits regardless of the university.
- Selected as one of 20 prospective students nationwide in Physics.

Merit-based Scholarship (Full Tuition Waiver, 3 Semesters)

Fall 2019 – Fall 2020

- Granted to outstanding students, covering full tuition fees; awarded only to top achievers.

Dean's List (4 Semesters)

- Awarded to top-performing students for exceptional academic excellence.

Extracurricular Experience

GLEAP

Mar 2021 – Present

Accredited Group of Outstanding Undergraduate Students
College of Natural Sciences, Seoul National University

- Collaborated with the education offices of Seoul and Busan to give natural science lectures to high school students.
- Engaged in intellectual exchanges not only within several members in this group but also with other institutions.

Language & Skills

Languages: Korean (Native), English (Intermediate)

Skills: Mathematica, $\text{\LaTeX}$, Python, MATLAB